

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4
Marks:40

Total

Annexure A

CASE STUDY

Code of Conduct:

I N. Chawran Aryan (192365053) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

N. Chawran Aryan

Signature of the student:

Case Study 2 — Autonomous Traffic Management Agent

Understanding the Case

A smart city traffic management system uses a First-Order Logic (FOL) Knowledge Base to automatically control traffic signals whenever an emergency vehicle such as an ambulance, fire engine, or police vehicle is detected.

The objective of the system is to:

- Identify emergency vehicles.
- Provide a clear path for emergency vehicles.
- Turn traffic signals green for emergency vehicles.
- Allow vehicles behind the emergency vehicle to move safely.
- Ensure quick and safe transportation during emergencies.

(a) Apply Unification to match Rule 1 with the given facts

Rule 1

[
forall x:\ Vehicle(x)\ \land\ EmergencyType(x)\ \rightarrow\ ClearPath(x)
]

Given Facts

- Vehicle(Ambulance)
- EmergencyType(Ambulance)

Step 1: Match Vehicle(x) with Vehicle(Ambulance)

Predicate:

- Vehicle(x)
- Vehicle(Ambulance)

Substitution:

[
\theta_1 = {x/Ambulance}
]

Step 2: Match EmergencyType(x) with EmergencyType(Ambulance)

Apply θ_1 :

- EmergencyType(Ambulance)
- EmergencyType(Ambulance)

Substitution:

[
 $\theta_2 = \{x/\text{Ambulance}\}$
]

(No new substitution is required.)

Most General Unifier (MGU)

[
 $\theta = \{x/\text{Ambulance}\}$
]

Applying Rule 1

Vehicle(Ambulance) $\wedge$ EmergencyType(Ambulance)

$\Rightarrow$

[
 ClearPath(Ambulance)
]

(b) Prove Proceed(CarA) using Resolution

Step 1: Convert rules into CNF

Rule 1

Vehicle(x) $\wedge$ EmergencyType(x) $\rightarrow$ ClearPath(x)

CNF:

[
 $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
]

Rule 2

$\text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

Split into two clauses.

For GreenSignal:

[
 $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$
]

For RedSignal:

[
 $\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$
]

Only the GreenSignal clause is needed.

Rule 3

$\text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y)$

$\rightarrow \text{Proceed}(x)$

CNF:

[
 $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$
]

Facts

1. $\text{Vehicle}(\text{Ambulance})$
2. $\text{EmergencyType}(\text{Ambulance})$
3. $\text{Vehicle}(\text{CarA})$
4. $\text{Behind}(\text{CarA}, \text{Ambulance})$

Negated Goal

Goal:

[
 $\neg \text{Proceed}(\text{CarA})$
]

Negate it:

[
 $\boxed{\neg \text{Proceed}(\text{CarA})}$
]

Resolution Steps

Step 1

Clause:

[
 $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
]

Resolve with

$\text{Vehicle}(\text{Ambulance})$

Substitution:

[
 $x = \text{Ambulance}$
]

Result:

[
 $\neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$
]

Resolve with

$\text{EmergencyType}(\text{Ambulance})$

Result:

[
 $\boxed{\text{ClearPath}(\text{Ambulance})}$
]

Step 2

Clause:

[
 $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$
]

Resolve with

$\text{ClearPath}(\text{Ambulance})$

Substitution:

[
 $x = \text{Ambulance}$
]

Result:

[
 $\boxed{\text{GreenSignal}(\text{Ambulance})}$
]

Step 3

Clause:

[
 $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x, y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$
]

Substitute

[
 $x = \text{CarA}, \quad y = \text{Ambulance}$
]

Result:

[
 $\neg \text{Vehicle}(\text{CarA}) \vee$
 $\neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee$
 $\neg \text{GreenSignal}(\text{Ambulance}) \vee$
 $\text{Proceed}(\text{CarA})$
]

Resolve with

$\text{Vehicle}(\text{CarA})$

Result:

$$[\neg \text{Behind}(\text{CarA}, \text{Ambulance}) \vee \neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})]$$

Resolve with

$$\text{Behind}(\text{CarA}, \text{Ambulance})$$

Result:

$$[\neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})]$$

Resolve with

$$\text{GreenSignal}(\text{Ambulance})$$

Result:

$$[\boxed{\text{Proceed}(\text{CarA})}]$$

Step 4 (Proof by Contradiction)

Resolve

$$\text{Proceed}(\text{CarA})$$

with

$$\neg \text{Proceed}(\text{CarA})$$

Result:

$$[\boxed{\square}]$$

(Empty clause)

Hence,

$$[\boxed{\text{Proceed}(\text{CarA}) \text{ is proved true.} }]$$

(c) Is the current FOL knowledge base sufficient for two simultaneous emergency vehicles?

No.

The current knowledge base is designed for only one emergency vehicle at a time. If two emergency vehicles (e.g., an ambulance and a fire truck) arrive simultaneously from different directions, the system cannot determine which one should receive priority or how to avoid conflicting green signals.

Limitations

1. No priority mechanism for emergency vehicles.
2. No conflict resolution at intersections.
3. No coordination between multiple traffic signals.
4. No rule to prevent two intersecting routes from receiving green signals simultaneously.
5. No mechanism to handle waiting emergency vehicles after the first one passes.

Additional Rules Needed

Rule 4: Priority Rule

If two emergency vehicles arrive simultaneously, the higher-priority vehicle gets precedence.

[
Emergency(x) ∧ Emergency(y) ∧ Priority(x) > Priority(y)
→ FirstPass(x)
]

Rule 5: Conflict Resolution

If two routes intersect, only one signal can be green.

[
Intersect(Route(x), Route(y))
→
¬(GreenSignal(x) ∧ GreenSignal(y))
]

Rule 6: Sequential Clearance

After one emergency vehicle passes, the next emergency vehicle receives a green signal.

[
Passed(x)\land Waiting(y)
\rightarrow GreenSignal(y)
]

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Case Study 3 — Agricultural Expert Reasoning System

Understanding the Case

An **Agricultural Expert Reasoning System** is an AI-based advisory system that assists farmers in making informed decisions about irrigation and crop management using **Propositional Logic**. The system stores agricultural knowledge as logical rules and facts and applies logical reasoning to recommend suitable farming practices.

In this case, the system determines:

- Whether irrigation is required.
- Whether the drip irrigation method should be applied.
- Whether the crop is at risk if irrigation is not applied.

Given :

P1: SoilDry $\rightarrow$ IrrigationNeeded

P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

P3: $\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk

Facts:

- SoilDry = True
- CropWheat = True

(a) Convert all sentences into Conjunctive Normal Form (CNF)

P1: SoilDry $\rightarrow$ IrrigationNeeded

Step 1: Eliminate implication

[
SoilDry $\rightarrow$ IrrigationNeeded
]

becomes

[
 $\neg$ SoilDry $\vee$ IrrigationNeeded
]

Step 2: Move $\neg$ inward

Already done.

Step 3: Distribute

No distribution needed.

CNF

[
\boxed{\neg SoilDry \vee IrrigationNeeded}
]

P2: $IrrigationNeeded \wedge CropWheat \rightarrow ApplyDripMethod$

Step 1: Eliminate implication

[
\neg(IrrigationNeeded \wedge CropWheat) \vee ApplyDripMethod
]

Step 2: Move $\neg$ inward (De Morgan's Law)

[
(\neg IrrigationNeeded \vee \neg CropWheat) \vee ApplyDripMethod
]

Step 3: Distribute

No further distribution required.

CNF

[
\boxed{\neg IrrigationNeeded \vee \neg CropWheat \vee ApplyDripMethod}
]

P3: $\neg ApplyDripMethod \wedge CropWheat \rightarrow CropAtRisk$

Step 1: Eliminate implication

[
\neg(\neg ApplyDripMethod \wedge CropWheat) \vee CropAtRisk
]

Step 2: Move $\neg$ inward

[
ApplyDripMethod \vee \neg CropWheat \vee CropAtRisk
]

Step 3: Distribute

No distribution required.

CNF

```
[  
\boxed{ApplyDripMethod \vee \neg CropWheat \vee CropAtRisk}  
]
```

Facts

SoilDry = True

CNF:

```
[  
\boxed{SoilDry}  
]
```

CropWheat = True

CNF:

```
[  
\boxed{CropWheat}  
]
```

Final CNF Clauses

C1: ($\neg$ SoilDry $\vee$ IrrigationNeeded)

C2: ($\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod)

C3: (ApplyDripMethod $\vee$ $\neg$ CropWheat $\vee$ CropAtRisk)

C4: (SoilDry)

C5: (CropWheat)

(b) Prove ApplyDripMethod using Resolution Refutation

Goal

Prove

[
ApplyDripMethod
]

Negate the Goal

[
\boxed{\neg ApplyDripMethod}
]

Add it to the KB.

Resolution Steps

Clauses

C1: ($\neg$ SoilDry $\vee$ IrrigationNeeded)

C2: ($\neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod)

C4: (SoilDry)

C5: (CropWheat)

C6: ($\neg$ ApplyDripMethod)

Step 1

Resolve

C1:

[
 $\neg$ SoilDry $\vee$ IrrigationNeeded
]

with

C4:

[
SoilDry
]

Result:

[
\boxed{IrrigationNeeded}
]

Step 2

Resolve

C2:

[
\neg IrrigationNeeded\vee\neg CropWheat\vee ApplyDripMethod
]

with

IrrigationNeeded

Result:

[
\neg CropWheat\vee ApplyDripMethod
]

Resolve with

CropWheat

Result:

[
\boxed{ApplyDripMethod}
]

Step 3

Resolve

ApplyDripMethod

with

[
\neg ApplyDripMethod
]

Result:

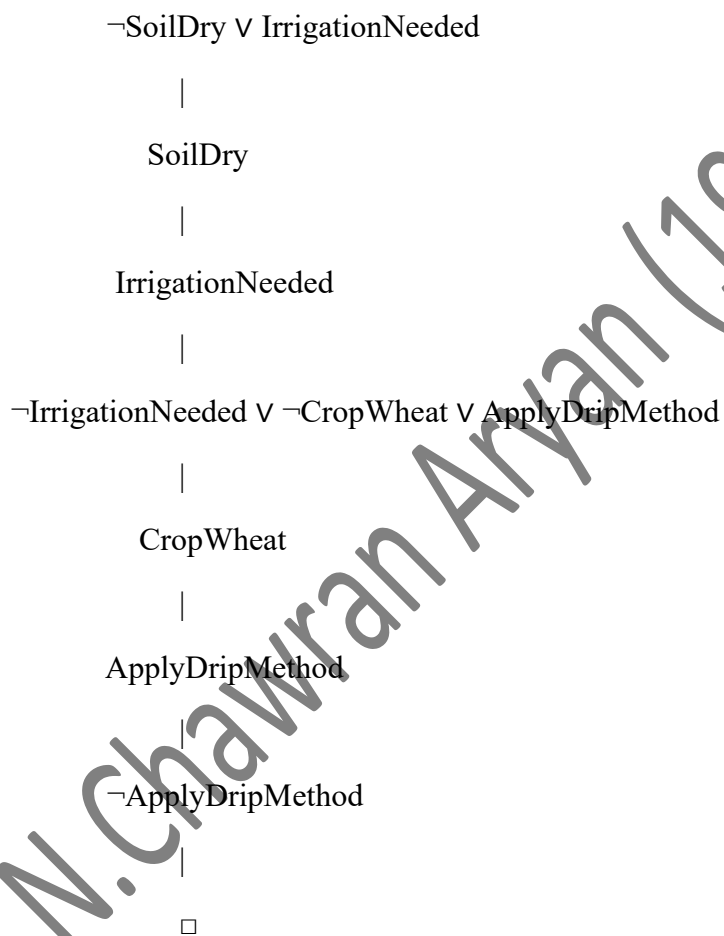
[
 $\boxed{\square}$
]

(Empty Clause)

Therefore,

[
 $\boxed{\text{ApplyDripMethod} \text{ is proved true.}}$
]

Resolution Proof Tree



(c) If 'SoilDry' is removed from the KB

New Knowledge Base

P1: SoilDry $\rightarrow$ IrrigationNeeded

P2: IrrigationNeeded $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

P3: $\neg$ ApplyDripMethod $\wedge$ CropWheat $\rightarrow$ CropAtRisk

Fact:

- CropWheat = True

(SoilDry is removed.)

Step 1

Can we infer IrrigationNeeded?

Rule:

[
SoilDry $\rightarrow$ IrrigationNeeded
]

Since **SoilDry is not known**, Modus Ponens **cannot** be applied.

Therefore,

[
 $\boxed{\text{IrrigationNeeded} \text{ cannot be concluded.}}$
]

Step 2

Can we infer ApplyDripMethod?

Rule:

[
IrrigationNeeded $\wedge$ CropWheat
 $\rightarrow$ ApplyDripMethod
]

We know:

- CropWheat = True
- IrrigationNeeded = Unknown

Hence,

[
\boxed{\text{ApplyDripMethod}} \text{ cannot be concluded.}]

Step 3

Can we infer CropAtRisk?

Rule:

[
\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \\ \rightarrow \text{CropAtRisk}

We know only:

- CropWheat = True

We **do not know** that $\neg \text{ApplyDripMethod}$ is true. Failure to prove ApplyDripMethod does **not** imply $\neg \text{ApplyDripMethod}$ (under classical logic).

Therefore,

[
\boxed{\text{CropAtRisk}} \text{ cannot be concluded.}]

Reasoning Pattern Used

- **Modus Ponens:** Cannot be applied because the premise **SoilDry** is missing.
- **Resolution:** Cannot derive **ApplyDripMethod** without **IrrigationNeeded**.
- **Classical Logic Principle:** The absence of proof for **ApplyDripMethod** does **not** mean $\neg \text{ApplyDripMethod}$ is true.